

HOUGHTON COUNTY AP, MI, USA

WMO: 727440

Lat: 47.169N

Lon: 88.489W

Elev: 1095

StdP: 14.12

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14858

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.1	-1.1	-12.7	2.9	-5.2	-8.1	3.7	-0.5	32.4	21.0	28.6	20.5	10.4	310	0.669

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	84.1	70.0	81.0	67.8	78.0	66.1	72.3	81.2	70.1	78.0	68.1	75.3	10.8	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.2	111.9	77.8	67.1	103.8	74.9	65.2	97.0	72.9	36.6	81.1	34.7	78.0	33.0	75.5	81.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
27.3	24.1	20.4	-11.6	91.2	6.0	3.0	-15.9	93.3	-19.4	95.1	-22.8	96.8	-27.2	99.0		
			-12.1	76.3	5.9	2.1	-16.3	77.8	-19.8	79.0	-23.1	80.2	-27.4	81.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.4	16.4	17.4	25.7	37.6	50.3	60.1	65.5	64.4	57.3	45.0	32.6	22.3
	DBStd	19.64	9.81	10.57	11.24	8.94	8.68	7.43	6.37	5.96	8.27	8.49	8.97	9.05
	HDD50	4823	1042	914	756	385	102	8	0	0	25	208	524	859
	HDD65	8857	1507	1334	1218	823	462	183	73	84	255	623	971	1324
	CDD50	1665	0	0	3	12	112	309	482	448	245	51	3	0
	CDD65	224	0	0	0	0	8	34	89	66	25	2	0	0
	CDH74	1697	0	0	1	3	95	305	668	449	164	12	0	0
	CDH80	370	0	0	0	0	23	84	142	88	32	1	0	0
Wind	WSAvg	9.7	10.7	10.8	10.0	10.1	9.7	8.7	8.6	8.4	8.8	10.0	10.2	10.2
Precipitation	PrecAvg	31.7	3.5	1.9	2.0	1.9	2.6	2.7	2.8	2.7	3.1	2.6	2.6	2.9
	PrecMax	42.2	8.5	5.1	7.5	5.7	5.7	6.6	6.2	6.5	6.6	5.7	5.4	7.3
	PrecMin	17.2	0.2	0.1	0.2	0.3	0.5	0.6	0.6	0.2	1.1	1.0	0.7	0.1
	PrecStd	6.3	1.9	1.3	1.5	1.0	1.3	1.4	1.3	1.4	1.4	1.3	1.2	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.8	45.0	59.6	71.3	83.0	88.4	88.1	86.7	84.3	74.5	61.5	44.8
		MCWB	35.6	39.4	52.1	55.2	65.4	71.8	72.6	71.2	69.9	61.4	51.1	39.0
	2%	DB	35.6	39.3	50.5	62.8	75.4	81.5	83.6	82.3	78.7	67.8	54.0	39.6
		MCWB	33.3	35.1	43.7	49.1	60.8	66.7	70.3	69.4	66.8	57.9	48.0	36.5
	5%	DB	33.2	35.7	44.9	56.7	70.1	76.9	80.9	78.9	74.4	62.8	49.2	36.9
		MCWB	31.6	33.0	39.9	46.2	57.0	64.7	68.3	67.1	64.9	55.3	43.7	34.7
	10%	DB	30.4	32.0	40.5	51.4	65.2	73.2	77.8	75.8	70.5	57.7	45.1	34.5
		MCWB	28.8	30.1	36.1	42.8	54.5	62.4	66.5	65.4	62.6	52.1	41.5	32.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.2	39.9	54.2	57.6	67.9	74.9	75.3	74.5	71.7	64.0	53.7	41.0
		MCDB	38.7	43.9	58.8	67.3	77.0	86.0	84.5	82.1	80.7	71.2	58.6	43.1
	2%	WB	33.7	35.9	44.8	51.0	63.0	70.2	72.4	71.4	68.8	59.7	49.1	37.2
		MCDB	35.3	38.6	49.1	60.0	72.4	78.5	81.1	79.9	76.0	65.6	54.0	38.7
	5%	WB	31.7	32.8	40.2	46.5	59.4	67.0	70.1	69.1	66.4	56.0	44.5	34.9
		MCDB	33.1	35.2	44.5	55.6	67.7	74.3	78.1	76.0	72.6	61.2	48.0	36.8
	10%	WB	28.8	30.1	36.1	43.3	55.5	64.1	68.0	67.1	63.7	52.7	41.7	32.7
		MCDB	30.4	32.0	39.9	50.5	63.6	70.7	75.5	73.9	69.2	57.1	45.0	34.5
Mean Daily Temperature Range	5% DB	MDBR	10.7	13.3	15.1	15.5	19.0	19.7	19.0	18.1	16.9	13.5	10.7	10.2
		MCDBR	12.3	16.6	18.8	25.0	26.6	24.2	22.1	22.0	21.8	21.0	17.5	12.6
		MCWBR	11.2	13.4	13.6	15.8	15.0	13.8	12.2	11.6	11.9	13.5	12.3	10.7
	5% WB	MCDBR	11.7	15.0	17.7	23.1	24.4	22.3	20.2	19.3	19.4	18.6	16.0	11.6
		MCWBR	11.0	12.5	13.4	15.7	15.4	14.4	12.3	11.4	12.0	13.1	12.2	10.4
Clear-Sky Solar Irradiance	taub	0.259	0.278	0.302	0.338	0.367	0.374	0.384	0.366	0.356	0.321	0.291	0.260	
	taud	2.451	2.378	2.389	2.377	2.386	2.381	2.408	2.456	2.468	2.535	2.534	2.486	
	Ebn at Noon	270	285	293	291	285	282	278	279	271	267	255	256	
	Edh at Noon	20	27	32	36	36	37	36	33	30	23	19	18	
All-Sky Solar Radiation	RadAvg	328	630	1093	1469	1744	1900	1940	1651	1207	717	366	239	
	RadStd	32	75	104	162	126	130	87	108	66	85	53	29	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (24 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page